

ROHIT MAITY

BACKEND / FULL-STACK DEVELOPER

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Third-year CSE student who designs and ships **production, distributed full-stack systems** — three deployed apps spanning real-time WebSockets, WebRTC, multimodal RAG and a private science engine. Strongest in **backend architecture, API security and system design**. Smart India Hackathon 2025 National Finalist.

SELECTED PROJECTS

SkyGuide AI

[GitHub](#) · [Live](#)

Real-time celestial recommendation & telescope-alignment platform · React 19 · Express 5 · FastAPI · Socket.IO · MongoDB

- Architected a **3-service distributed system** — React SPA, Express gateway, and a private FastAPI science engine — with strict responsibility boundaries and server-to-server auth over a shared internal key.
- Built live telescope alignment: phone orientation streams over WebSocket, guiding users onto any of **13,311 catalogued deep-sky objects**.
- Hardened auth end-to-end — HTTP-only JWT cookies, bcrypt, and layered rate limiting (login / OTP / password-reset / LLM-spend).

Neuron

[GitHub](#) · [self-hosted demo](#)

Multimodal semantic search & grounded document chat · FastAPI · CLIP · Whisper · FAISS · React 19

- Designed a layered FastAPI backend fusing text, image and audio into a **single CLIP query vector**, re-ranked by visual similarity blended with dense-embedding and BM25 signals.
- Built location-aware RAG so every answer carries clickable citations back to the exact source page, heading and line range.
- Engineered **graceful degradation** — lazy model loading with Groq → local → extractive fallbacks keep the API serving when optional ML deps are absent.

Yap Chat

[GitHub](#) · [Live](#)

Code-based chat rooms with P2P group video calling · MERN · Socket.IO · WebRTC

- Implemented **peer-to-peer group video for up to 8** over a WebRTC mesh with Socket.IO signalling — screen share, host handover and audio-only fallback.
- Secured the realtime layer with JWT-authenticated socket handshakes, NoSQL-injection coercion and per-message authorization.
- Verified against a live server with **35 passing tests** (20/20 call signalling, 15/15 API security).

CodeAtlas AI · in progress

AI-assisted codebase exploration & navigation tool — architecture and backend in active development.

SKILLS

LANGUAGES

Python · Java (OOP) · JavaScript · C

BACKEND

Node.js · Express · Socket.IO · FastAPI · REST · JWT auth · rate limiting

SYSTEMS

Microservices · distributed systems · system design · WebRTC / WebSocket realtime

DATABASES

MongoDB · MySQL · Redis

FRONTEND

React · Tailwind · HTML / CSS · Bootstrap

TOOLS

Git · Figma

ACHIEVEMENTS

SIH 2025 — National Finalist

Smart India Hackathon (PS 25007, Livestock/Agriculture). Contributed to *LivestockIQ* backend — Node/Express/MongoDB, JWT, PDFKit reporting for AMU/MRL monitoring.

SBI Ideathon 2025 — Top 1000 (National)

Technical Lead on *PraanaCare* (AI telemedicine) — owned technical architecture, stack selection and system design.

EDUCATION

Heritage Institute of Technology, Kolkata

B.Tech, Computer Science & Engineering (IoT, Cybersecurity & Blockchain) · 2024–2028

CGPA 9.02 / 10 · no backlogs

Class XII 90% · Class X 96% (State Rank 20)